CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): A sky-diver falls at constant speed after opening the parachute. Reason (R): The forces on her are balanced. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Dams are made thicker at the bottom. Reason (R): Water pressure increases with depth. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): A gas can be compressed easily but a solid cannot. Reason (R): Gas particles are widely separated while solid particles are tightly packed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Brass is a mixture of copper and zinc. Reason (R): The proportions of copper and zinc in brass can vary. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): A soft drink goes flat when left open. Reason (R): The solubility of a gas in a liquid decreases as pressure decreases. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A man weighs 600 N on Earth. On a planet where gravity is 1/3 of Earth's, his weight is: 1

(a) 300 N **(b)** 600 N **(c)** 1800 N **(d)** 200 N

7. A box exerts 2000 Pa. If its base area is halved without changing weight, the pressure becomes: 1

(a) 500 Pa **(b)** 1000 Pa **(c)** 4000 Pa **(d)** 2000 Pa

8. A sample of a pure solid is heated at a constant rate and the temperature is recorded. A horizontal portion on the graph shows: 1

(a) boiling only **(b)** change of state **(c)** chemical change **(d)** freezing only

9. 9 g of water is decomposed. The masses of hydrogen and oxygen obtained are: 1

(a) 4.5 g and 4.5 g **(b)** 8 g and 1 g **(c)** 1 g and 8 g **(d)** 2 g and 7 g

10. A solution is made by mixing 200 g of 10 per cent solution and 300 g of 20 per cent solution of the same solute. The percentage of the final solution is: 1

(a) 14 per cent **(b)** 15 per cent **(c)** 16 per cent **(d)** 30 per cent

11. A magnet is brought near an iron nail and the nail jumps to the magnet. The force that does the work is: 1

(a) electrostatic force **(b)** friction
(c) gravity **(d)** magnetic force

12. A ship at sea is advised to move away from the path of a cyclone because: 1

(a) the eye is cold
(b) the sea is calm
(c) wind and waves are strongest near the eye wall
(d) ships are heavy

13. In a closed bottle half-filled with water at constant temperature, the rate of evaporation compared with the rate of condensation is: 1

(a) less always **(b)** equal at equilibrium
(c) zero **(d)** greater always

14. Zinc reacts with dilute acid and produces: 1

(a) hydrogen gas **(b)** nitrogen gas **(c)** oxygen gas **(d)** chlorine gas

15. Alloys such as brass are solutions in which: 1

(a) a solid dissolves in a solid **(b)** a liquid in a liquid
(c) a gas in a gas **(d)** a gas in a liquid

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does a block slide down a slope faster when the slope is polished? 2

17. State why weather satellites are used for cyclone tracking. 2

18. Why does sweating not cool us effectively when the humidity is high? 2

19. Explain why alloys like solder are used to join metals. 2

20. Explain why boiled and cooled water tastes flat. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Compare ice, water and steam in a table in terms of energy, spacing and motion of particles. 3

22. Explain why diamonds do not conduct electricity but graphite does. 3

23. Explain how crystallisation purifies copper sulphate. 3

24. Why does an object dropped from a height of 1 m take less time to reach the ground than from 2 m? 3

OR

Explain why helium balloons rise but ordinary air balloons do not. 3

25. How would the boiling point change if you added salt to water? Explain briefly. 3

OR

A fruit juice is made of water, sugar, acids and colour. Describe its nature and how you may separate water from it. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The solubility of a salt is 20 g per 100 g of water at 20 degrees Celsius and 50 g per 100 g of water at 70 degrees Celsius. 150 g of the saturated solution at 70 degrees Celsius is cooled to 20 degrees Celsius. Find the mass of solute at 70, mass of water, and mass of crystals formed. 5

OR

Discuss how the shape of a body affects air and water resistance and give five examples of streamlining in nature and technology. 5

27. A cylinder with a piston traps air. Explain what happens to the pressure when the piston is pushed in or pulled out, and relate it to a bicycle pump. 5

OR

Calculate: how many times does the volume increase when 1 mL of water changes to steam if the volume of steam is about 1600 mL at 100 degrees Celsius? Explain the reason. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND